

Department of Mathematics, IIT Madras
MA 2031 - Linear Algebra for Engineers

Quiz II

Answer all questions. Time: 50 minutes Max. marks: 20

1. (a) Find a least squares solution to the following system of equations:

$$3x + y = 1, \quad x + 2y = 0, \quad 2x - y = -2.$$

(3 marks)

- (b) Let $A \in \mathbb{C}^{m \times n}$ and $b \in \mathbb{C}^m$. When does the (possibly inconsistent) system $Ax = b$ have a *unique* least squares solution? Justify your answer.

(2 marks)

2. Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^4$ be defined by

$$T(a, b, c) = (a - b, b - c, a - c, a - 2b + c), \quad a, b, c \in \mathbb{R}.$$

Find bases for the range space and the null space of T .

(4 marks)

3. Let $T : V \rightarrow V$ be a linear map satisfying $T^2 = T$. Show that

(a) $V = R(T) + N(T)$

(b) $R(T) \cap N(T) = \{0\}$.

(2 + 2 marks)

4. Let $T : \mathbb{F}^n \rightarrow \mathbb{F}^n$ be defined by $T(z_1, z_2, \dots, z_n) = (0, z_1, \dots, z_{n-1})$. Obtain the formula for $T^*(w)$, $w \in \mathbb{F}^n$.

(3 marks)

5. Let V denote the vector space of all polynomials in the single variable t , with coefficients from $\mathbb{R}$. Suppose that P_n and P_{n+1} are the subspaces of V consisting of polynomials of degree at most n and $n + 1$, respectively.

(a) Let $T : P_n \rightarrow P_{n+1}$ be the linear map defined by $T(p(t)) = \int_0^t p(x) dx$.

Compute the nullity and the rank of T .

(b) Let $S : V \rightarrow V$ be the linear map defined by $S(p(t)) = \int_0^t p(x) dx$.

Is S an isomorphism? Justify your conclusion.

(2 + 2 marks)
